

IIT JAM 2006 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

15

TOTAL MARKS

90 M

TEST DURATION

60 Min

PAPER PATTERN

MCQ

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2006 • Q1 • ID: JAM_2006_Q1)

MCQ

+6 M

Neg: -2.0

1. $\lim_{n \rightarrow \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$ equals
- (A) 3
(B) 2
(C) 1
(D) 0

Question 2 (JAM 2006 • Q2 • ID: JAM_2006_Q2)

MCQ

+6 M

Neg: -2.0

2. Let $f(x) = (x-2)^{17}(x+5)^{24}$. Then
- (A) f does not have a critical point at 2
(B) f has a minimum at 2
(C) f has a maximum at 2
(D) f has neither a minimum nor a maximum at 2

Question 3 (JAM 2006 • Q3 • ID: JAM_2006_Q3)

MCQ

+6 M

Neg: -2.0

3. Let $f(x, y) = x^5 y^2 \tan^{-1}\left(\frac{y}{x}\right)$. Then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ equals
- (A) $2f$
(B) $3f$
(C) $5f$
(D) $7f$

Question 4 (JAM 2006 • Q4 • ID: JAM_2006_Q4)

MCQ

+6 M

Neg: -2.0

4. Let G be the set of all irrational numbers. The interior and the closure of G are denoted by G^0 and $\overline{G}$, respectively. Then
- (A) $G^0 = \emptyset$, $\overline{G} = G$
(B) $G^0 = \mathbb{R}$, $\overline{G} = \mathbb{R}$
(C) $G^0 = \emptyset$, $\overline{G} = \mathbb{R}$
(D) $G^0 = G$, $\overline{G} = \mathbb{R}$

Question 5 (JAM 2006 • Q5 • ID: JAM_2006_Q5)

MCQ

+6 M

Neg: -2.0

5. Let $f(x) = \int_{\sin x}^{\cos x} e^{-t^2} dt$. Then $f'(\pi/4)$ equals
- (A) $\sqrt{1/e}$
(B) $-\sqrt{2/e}$
(C) $\sqrt{2/e}$
(D) $-\sqrt{1/e}$

Question 6 (JAM 2006 • Q6 • ID: JAM_2006_Q6)

MCQ

+6 M

Neg: -2.0

6. Let C be the circle $x^2 + y^2 = 1$ taken in the anti-clockwise sense. Then the value of the integral
- $$\int_C \left[(2xy^3 + y) dx + (3x^2y^2 + 2x) dy \right]$$
- equals

- (A) 1
(B) $\pi/2$
(C) π
(D) 0

Question 7 (JAM 2006 • Q7 • ID: JAM_2006_Q7)

MCQ

+6 M

Neg: -2.0

7. Let r be the distance of a point $P(x, y, z)$ from the origin O . Then ∇r is a vector
- (A) orthogonal to $\overrightarrow{OP}$
(B) normal to the level surface of r at P
(C) normal to the surface of revolution generated by OP about x-axis
(D) normal to the surface of revolution generated by OP about y-axis

Question 8 (JAM 2006 • Q8 • ID: JAM_2006_Q8)

MCQ

+6 M

Neg: -2.0

8. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_2, 0).$$

If $N(T)$ and $R(T)$ denote the null space and the range space of T respectively, then

- (A) $\dim N(T) = 2$
- (B) $\dim R(T) = 2$
- (C) $R(T) = N(T)$
- (D) $N(T) \subset R(T)$

Question 9 (JAM 2006 • Q9 • ID: JAM_2006_Q9)

MCQ

+6 M

Neg: -2.0

9. Let S be a closed surface for which $\iint_S \vec{r} \cdot \hat{n} d\sigma = 1$. Then the volume enclosed by the surface is

- (A) 1
- (B) $1/3$
- (C) $2/3$
- (D) 3

Question 10 (JAM 2006 • Q10 • ID: JAM_2006_Q10)

MCQ

+6 M

Neg: -2.0

10. If $(c_1 + c_2 \ln x)/x$ is the general solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + kx \frac{dy}{dx} + y = 0, \quad x > 0,$$

then k equals

- (A) 3
- (B) -3
- (C) 2
- (D) -1

Question 11 (JAM 2006 • Q11 • ID: JAM_2006_Q11)

MCQ

+6 M

Neg: -2.0

11. If A and B are 3×3 real matrices such that $\text{rank}(AB)=1$, then $\text{rank}(BA)$ **cannot** be

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Question 12 (JAM 2006 • Q12 • ID: JAM_2006_Q12)

MCQ

+6 M

Neg: -2.0

12. The differential equation representing the family of circles touching y -axis at the origin is

- (A) linear and of first order
- (B) linear and of second order
- (C) nonlinear and of first order
- (D) nonlinear and of second order

Question 13 (JAM 2006 • Q13 • ID: JAM_2006_Q13)

MCQ

+6 M

Neg: -2.0

13. Let G be a group of order 7 and $\phi(x) = x^4$, $x \in G$. Then ϕ is

- (A) not one – one
- (B) not onto
- (C) not a homomorphism
- (D) one – one, onto and a homomorphism

Question 14 (JAM 2006 • Q14 • ID: JAM_2006_Q14)

MCQ

+6 M

Neg: -2.0

14. Let R be the ring of all 2×2 matrices with integer entries. Which of the following subsets of R is an integral domain?

- (A) $\left\{ \begin{pmatrix} 0 & x \\ y & 0 \end{pmatrix} : x, y \in \mathbf{Z} \right\}$
- (B) $\left\{ \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} : x, y \in \mathbf{Z} \right\}$
- (C) $\left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} : x \in \mathbf{Z} \right\}$
- (D) $\left\{ \begin{pmatrix} x & y \\ y & z \end{pmatrix} : x, y, z \in \mathbf{Z} \right\}$

Question 15 (JAM 2006 • Q15 • ID: JAM_2006_Q15)

MCQ

+6 M

Neg: -2.0

15. Let $f_n(x) = n \sin^{2n+1} x \cos x$. Then the value of

$$\lim_{n \rightarrow \infty} \int_0^{\pi/2} f_n(x) dx - \int_0^{\pi/2} \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx$$

is

- (A) $1/2$
- (B) 0
- (C) $-1/2$
- (D) $-\infty$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2006 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	A
2	MCQ	+6	D
3	MCQ	+6	D
4	MCQ	+6	C
5	MCQ	+6	B
6	MCQ	+6	C
7	MCQ	+6	B
8	MCQ	+6	A

Q. No.	Type	Marks	Official Key
9	MCQ	+6	B
10	MCQ	+6	A
11	MCQ	+6	D
12	MCQ	+6	C
13	MCQ	+6	D
14	MCQ	+6	C
15	MCQ	+6	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.